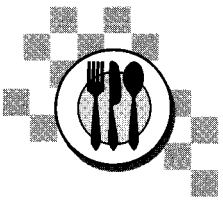
CHAPTER 12



# Meat and Poultry Sanitation

## About This Chapter

In this chapter you will learn:

1. Why it is so important to have good sanitation in meat- and poultry-processing facilities
2. Which pathogens cause most problems in meat and poultry food products
3. How to design and organize meat and poultry plants to control pathogens
4. How to clean and sanitize the various parts of meat- and poultry-processing plants.
5. How HACCP is important in meat- and poultry-processing plants

## Introduction

Meat and poultry are perishable foods (i.e., they spoil quickly), and the meat discolors easily. Poor sanitation allows microbes to cause even more damage to the flavor and color of meat. A good sanitation program reduces color changes and spoilage and increases the shelf life of meat and poultry.

In the meat and poultry industry, sanitation begins with the live animal or bird and continues until the food product is served. Management should plan, enforce, and supervise the sanitation program thoroughly. The person in charge of the program should report to top management and make sure that the facility and food products are sanitary. Trained staff should clean and inspect the plant and equipment and be responsible for keeping it clean.

## COMMON PATHOGENS

Meat and poultry products cause about 23% of foodborne illness outbreaks. They are associated with 5 to 10% of deaths caused by foodborne disease. Pathogens associated with meat and poultry are *Listeria monocytogenes, Escherichia coli* O157:H7, *Salmonella* spp., and *Campylobacter* spp.

*Listeria monocytogenes* is often found on retail cuts of fresh poultry and red meats. *L. monocytogenes* can also grow in some packaged cooked meat products, such as hotdogs. It is important to make sure that fresh meats used to make processed products do not recontaminate ready-to-eat products after they have been processed. Table 12.1 shows how often *L. monocytogenes* was found in various places in 41 meat plants.

**TABLE 12.1.** Percentage of Meat Plants That Tested Positive for *L. Monocytogenes* in Various Areas

| Location | Plants, % |
|---|---|
| Floors | 37 |
| Drains | 37 |
| Cleaning aids | 24 |
| Wash areas | 24 |
| Sausage peelers | 22 |
| Food contact surfaces | 20 |
| Condensation | 7 |
| Walls and ceilings | 5 |
| Compressed air | 4 |

*Source:* Wilson, G. D. 1987. Guidelines for production of ready-to-eat meat products. *Proc. Meat Ind. Res. Conf.*, p. 62. American Meat Institute, Washington, D.C.

*L. monocytogenes* is often found around wet areas and cleaning equipment, such as floors, drains, wash areas, ceiling condensation, mops and sponges, brine chillers, and at peeler stations. Refrigeration at 4 to 5°C (39–41°F) (a common storage temperature) does not stop this pathogen from growing. Possible ways to control *L. monocytogenes* include antimicrobial agents, colder storage (<2°C/<35°F), making products with lower water activity or pH, or pasteurizing products after processing.

One of the best ways to control pathogens like *L. monocytogenes* is by preventing cross-contamination. Staff who work in areas that contain both raw and finished product, such as smokehouses and water- and steam-cooking areas, should change

their outer clothing and sanitize their hands or change gloves when moving from raw- to finished-product areas. Staff should sanitize utensils and thermometers that are used for raw and finished products each time they use them. Staff must also scrub floors frequently. If liquid condenses on the ceiling, staff should remove it using a vacuum cleaner or sanitized sponge mop.

## SANITATION MANAGEMENT

Meat and poultry are ideal food for microorganisms that cause discoloration, spoilage, and foodborne illness. Good sanitation keeps contamination to a minimum and increases the stability of the product.

Some of the most important reasons for having high standards of cleanliness in meat and poultry facilities are:

- When conditions are unsanitary, microorganisms that attack meat and poultry are present. These microorganisms change the color and flavor of products.
- Self-service packages of fresh meat and poultry in stores make sanitation very important to extend the shelf life as long as possible.
- Good sanitary conditions reduce the amount of waste, because less-discolored or spoiled meat and poultry have to be thrown out.
- Excellent sanitary conditions can improve the image and reputation of a firm. A sanitary product looks and tastes better and is more healthful than tainted food.
- Regulatory agencies and consumers insist on good sanitation.
- Employees deserve clean, safe working conditions. Sanitary and tidy surroundings improve morale and productivity.
- More centralized processing and packaging requires more emphasis on sanitation. More processing and handling of food require a stricter sanitation program.
- Sanitation is good business.

Microorganisms are a major cause of color changes in meat and poultry. Microbes take up oxygen from the surface of the meat, so there is less oxygen to keep the muscle color bright. Different microorganisms have different effects on color pigments. But good cleanliness can slow the growth of microbes. Employees who handle meat should try to minimize the number of bacteria that first come in contact with the meat.

### Meat and Poultry Contamination

During slaughter, processing, distribution, and foodservice, meat and poultry items may be handled as many as 18 to 20 times. Almost anything that touches meat and poultry can contaminate it, so the risk of contamination increases with the number of times the product is handled.

A healthy, live animal has defenses (immune system or resistance) that stop bacteria from getting into its muscles. After slaughter, the natural defenses stop working, and there is a race between human beings and the microbes to see which will eat the meat! If processors handle the meat carelessly, the microbes win. Sanitation employ-

ees have to find ways to stop the microorganisms from growing. (Chapter 3 discusses contamination sources during slaughter and processing.)

### Layout and Plant Design

1. The plant layout should keep pests out and stop *L. monocytogenes* from moving between raw- and cooked-product areas. Designers need to consider employee traffic patterns, where support and supervisory staff work, and where and how food is handled.
2. Designers should use air and refrigeration equipment that is easy to clean and sanitize. Air should not flow from raw-food areas to cooked-food areas.
3. All equipment and other surfaces should have smooth, nonporous surfaces that are easy to clean and sanitize.
4. Floor surfaces should be easy to clean and should not collect water.
5. The design should prevent condensation from collecting.

### Equipment Design

1. The facility should use only regulatory approved materials.
2. Staff should maintain and store all equipment properly.

### Process Control

1. If the process does not contain a step to kill *L. monocytogenes,* the operation should be designed to minimize contamination.
2. The step to kill *L. monocytogenes* (if there is one) should be a critical control point in the Hazard Analysis Critical Control Point (HACCP) program.

### Operation Practices

1. Management should educate employees about their responsibility for using good manufacturing practices (GMPs) and HACCP.
2. Employees should have the right sanitation equipment, such as footbaths, hand dips, hairnets, and gloves.
3. Contamination sources should be removed, especially in areas where food is processed and ready to eat.
4. Management should understand and support GMPs and HACCP.

### Sanitation Practices

1. The operation should have an adequate number of employees, amount of time, and number of supervisors for cleaning and sanitizing.
2. The facility should have written cleaning and sanitizing procedures posted for each area.

3. The environment (equipment, walls, water, air, etc.) should be sampled to make sure that cleaning and sanitizing procedures are working.

## Checking Control of Pathogens

1. Weekly samples should be taken from plant areas, equipment, and the air supply for microbial counts. It is especially important to take samples from points between the step used to kill *L. monocytogenes* and packaging.
2. Composite (combined) samples are cheaper to analyze than individual samples. If a composite sample shows pathogens, sanitation staff need to go back and analyze the individual samples to find out which one is contaminated.

The following steps can help control pathogens:

1. Mechanically or manually scrub floors and drains every day. Rinse drains with disinfectant every day.
2. Scrub walls each week.
3. Clean the outside of all equipment, light fixtures, sills and ledges, piping, vents, and other processing and packaging areas that are not part of the daily cleaning program each week.
4. Clean cooling and heating units and ducts each week.
5. Caulk all cracks in walls, ceilings, and window and door sills.
6. Scrub and clean raw-material areas at least as often as the processing and packaging areas.
7. Keep hallways and passageways used by both raw and finished product clean and dry.
8. Minimize traffic in and out of processing and packaging areas. Establish plant traffic patterns to reduce cross-contamination from feet, containers, pallet jacks, pallets, and fork trucks.
9. Change outer clothing and sanitize hands or change gloves when moving from an area containing raw ingredients to an area containing finished food products.
10. Change into clean work clothes each day at the plant. Color-code clothing to show which clothes belong in each area.
11. Make sure visitors change into clean clothes at the plant.
12. Monitor the plant environment (surfaces and air) to make sure that *Listeria* are under control.
13. Close-in processing and packaging rooms so that they only receive filtered air.
14. Clean and sanitize all equipment and containers before bringing them into processing and packaging areas.

## Temperature Control

Meat and poultry spoil at high temperatures. Chemical and biochemical reactions are faster and the lag phase of the growth pattern of microorganisms is shorter at higher temperatures. Microbial and nonmicrobial spoilage both increase at higher tempera-

tures until 45°C (113°F). Above 60°C (140°F), microbes do not usually spoil food. (See Chap. 2 for more about microbial growth.)

Microorganisms grow fastest between 2°C and 60°C (36–140°F). This temperature range is the critical zone, or the danger zone. Meat and poultry must be stored either below or above this temperature zone and should be taken through this range as quickly as possible when a temperature change is necessary (e.g., cooking or chilling). Storage at temperatures below the critical zone (refrigerator or freezer) does not kill bacteria, but they can only grow and multiply very slowly. Below the critical zone, bacteria are less active and some may die through stress (see Fig. 12.1).

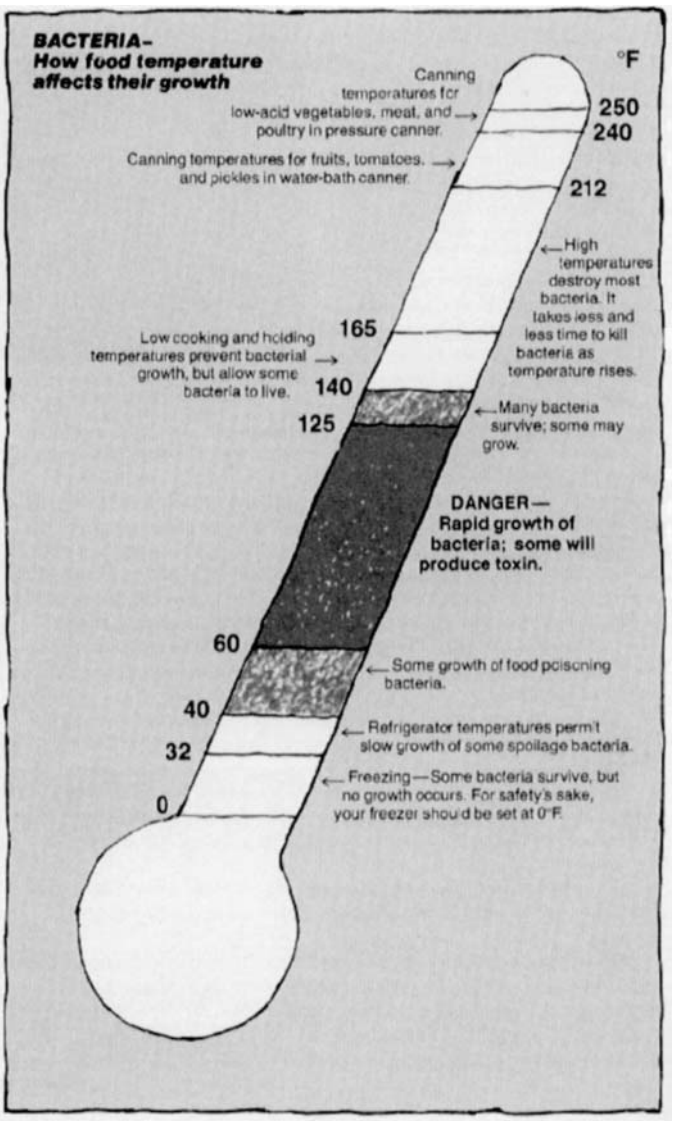



**FIGURE 12.1.** Bacteria—how food temperature affects their growth. (*Source: The Safe Food Book: Your Kitchen Guide,* p.8, Home & Garden Bulletin No. 241, United States Department of Agriculture, Food Safety and Inspection Service, July 1984.)

Processing and storage at colder temperatures reduces spoilage and also reduces growth of microorganisms on equipment, supplies, or other materials in the storage and processing areas. If conditions are unsanitary and the temperature is not controlled, some types of *Pseudomonas* can double in number every 20 minutes. This means that in 12 hours, 1 bacterium can become 280 trillion bacteria. At 0°C (32°F), meat and poultry have twice as long a shelf life as meat and poultry at 4.4°C (40°F) and at least 4 times as long a shelf life as meat and poultry at 10°C (50°F).

Air curtains can prevent cold air from leaving an area, especially if truck doors have to be left open, and keep out insects and dust. The air stream should go across the entire width of staff entrances.

It is important to clean the roofs above food-manufacturing areas. Equipment and exhaust stacks may be vented through the roof. If possible, designers should enclose processing equipment that hangs from the ceiling with a floor to separate it from the processing area. Particles can deposit on the outside of the roof, especially if it is flat, and can attract birds, rodents, or insects. Pools of water also encourage pests. The slope of the roof should be at least 1% for drainage.

## CLEANING PRACTICES FOR MEAT AND POULTRY PLANTS

An efficient cleaning system can reduce labor costs up to 50%. Proper construction and cleaning equipment are critical for a good cleaning operation. Floors, walls, and ceilings should be impervious (nonporous) and easily cleaned. Floors should have a slope of at least 10.5 mm/m (0.4 in/yd).

### Hot-Water Wash

Soil from meat and poultry is mostly fat and protein, so a hot-water wash does not work well. Hot water can loosen and melt fat, but it also tends to cook fats and proteins onto the surface. The main advantage of a hot-water-wash system is that it does not need much equipment. But it uses a lot of labor and causes water condensation on equipment, walls, and ceilings. It is difficult to remove heavy soil using a hot-water wash.

### High-Pressure, Low-Volume Cleaning

High-pressure, low-volume spray cleaning works well in the meat and poultry industry because it removes soil well. With this equipment, the operator can clean difficult-to-reach areas with less labor, and the cleaning compound works better at a cooler temperature.

Portable high-pressure, low-volume units can be easily moved around the plant to clean parts of equipment and building surfaces. It works especially well for conveyors and processing equipment where soaking is impractical and hand brushing is difficult and labor intensive.

Chapter 7 includes illustrations of the metering device and controls of a centralized high-pressure cleaning unit. Figure 12.2 shows a dispensing nozzle for this equipment.

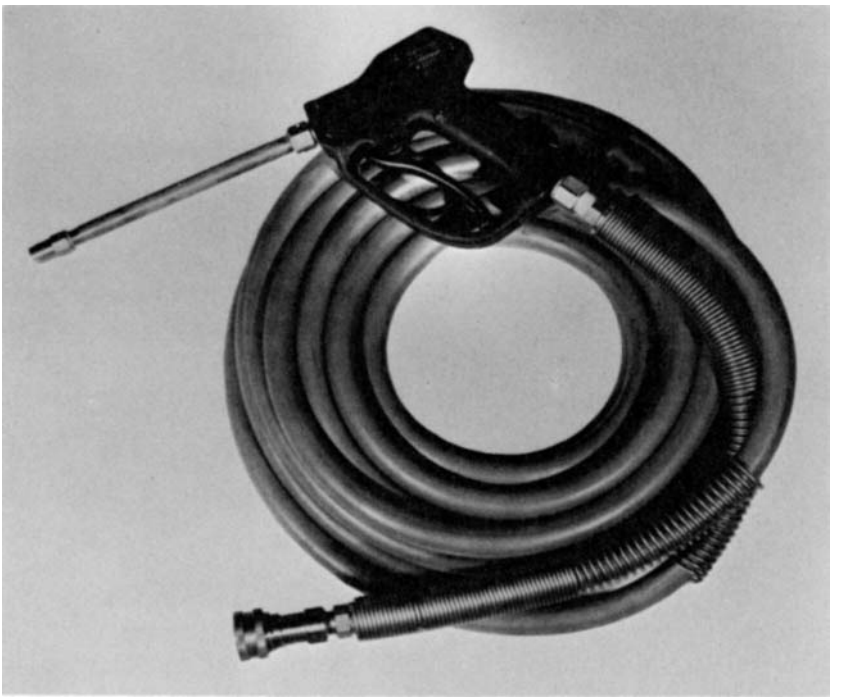


**FIGURE 12.2.** High-pressure hose with a female, stainless-steel, quick-connect, heavy-duty, dead-man shutoff-type spray gun, extension wand, and 4- to 12-liter-per-minute (1–3 gallon-per-minute) capacity spray nozzle. *(Courtesy of Klenzade Division, a service of Ecolab.)*

A centralized high-pressure cleaning system has many uses and benefits:

- The equipment can dispense several types of cleaning compound—acid, alkaline, or neutral cleaners and sanitizers.
- The system can use mechanized spray heads, mounted on belt conveyors with automatic washing, rinsing, and cutoff.
- Quick connection outlets are available in all areas requiring cleaning.

Centralized equipment is custom built and is far more expensive than portable units, depending on the size of the facility and the flexibility of the equipment. The initial cost may range from $15,000 to over $150,000.

## Foam Cleaning

Foam is useful for cleaning large surface areas of meat and poultry plants, the outside of transportation equipment, ceilings, walls, piping, belts, and storage containers. Portable foam equipment is similar in size and cost to portable high-pressure units. Centralized foam cleaning has the same benefits as other centralized systems.

## Gel Cleaning

This equipment is similar to high-pressure units, except that the cleaning compound is applied as a gel rather than as a high-pressure spray. Gel is good for cleaning packaging equipment because it clings to soiled surfaces. The equipment costs about the same as portable high-pressure units.

## Combination Centralized High-Pressure, Low-Volume, and Foam Cleaning

This system is the same as a centralized high-pressure system, except that the system can also apply foam. This method is flexible because foam can clean large surfaces and high pressure can clean belts, conveyors, and hard-to-reach areas. This system costs from $15,000 to over $150,000, depending on the size of the system.

## Cleaning in Place (CIP)

Nozzles inside the equipment apply a recirculating cleaning solution to automatically clean, rinse, and sanitize the equipment. The benefits of CIP systems are discussed in Chapter 7. CIP cannot be used much in the meat and poultry industry. The equipment is expensive and does not work well in heavily soiled areas. CIP cleaning is sometimes used to clean vacuum thawing chambers, pumping and brine circulation lines, silos, and fat-rendering systems. Figure 12.3 illustrates a CIP system for

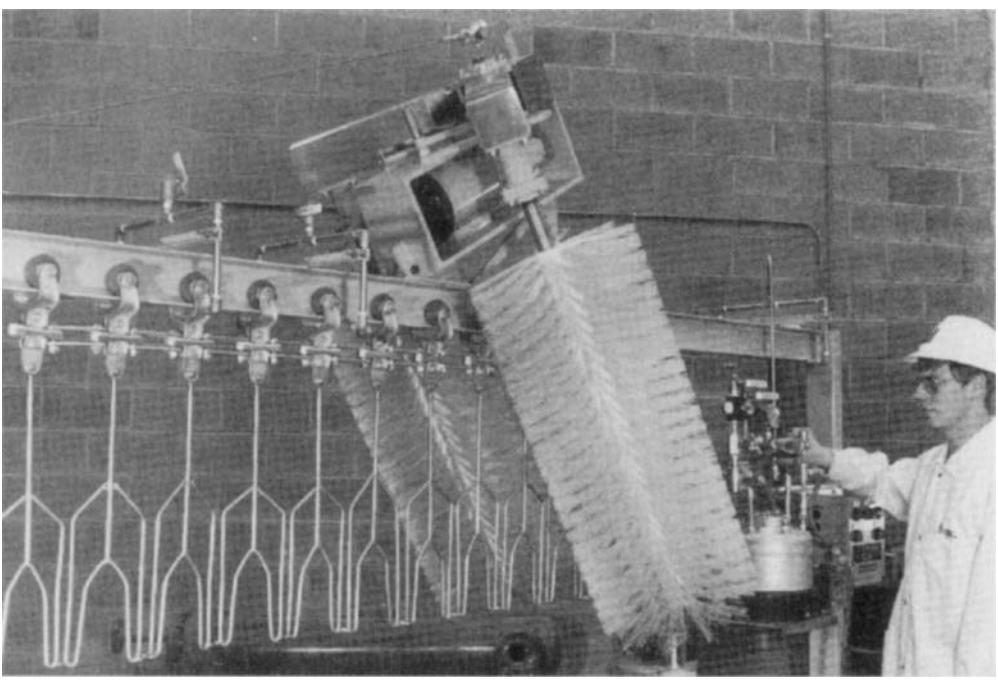


**FIGURE 12.3.** Shackle washer for cleaning shackles, rollers, and the chain in poultry-processing plants. *(Courtesy of Chemidyne Corp.)*

washing shackles, rollers, and the chain in poultry plants. The motor and drive components are mounted on a base plate. The shackles are cleaned as they pass between two rotating brushes. The brushes can be lifted above the rail when not in use.

## CLEANING COMPOUNDS FOR MEAT AND POULTRY PLANTS

### Acid Cleaners

Information about strongly and mildly acid cleaners is provided in Chapter 5.

### Strongly Alkaline Cleaners

Examples of strongly alkaline compounds are sodium hydroxide (caustic soda) and silicates with high ratios of nitrous oxide to silicon dioxide. Silicates make sodium hydroxide less corrosive and better at penetrating and rising. These cleaners remove heavy soils, such as those found in smokehouses.

### Heavy-Duty Alkaline Cleaners

These cleaners are frequently used with CIP, high-pressure, and other mechanized systems in meat and poultry plants.

### Mild Alkaline Cleaners

Mild cleaners are often liquid solutions and are used for hand cleaning lightly soiled areas in meat and poultry plants.

### Neutral Cleaners

Information about these and other cleaning compounds is given in Chapter 5.

## SANITIZERS FOR MEAT AND POULTRY PLANTS

Sanitizers work only on clean surfaces, i.e., after all dirt is removed. Soil that can prevent the sanitizer from working properly includes fats, meat juices, blood, grease, oil, and mineral buildup. Microbes can grow under and inside soil and can hold food and water that allow microbes to grow. Chemical sanitizers cannot destroy microorganisms in or under soil deposits.

### Steam

Steam is not a good sanitizer. Many operators think that water vapor is steam and do not expose the equipment to enough steam to sanitize it. Workers should not use steam to sanitize refrigerated areas because it causes condensation and wastes energy. Steam also does not sanitize conveyors properly.

### Chemical Sanitizers

Chlorine is the most important compound for disinfecting, sterilizing, and sanitizing equipment, utensils, and water. Meat and poultry operations most often use these chlorine compounds:

*Sodium and calcium hypochlorite:* These cost more than plain chlorine, but are easier to apply.

*Liquid chlorine:* This chlorinates processing and cooling waters and prevents bacterial slimes.

*Chlorine dioxide:* This works well to destroy bacteria when organic matter (such as food waste) is present because it does not react with protein.

*Active iodine* solutions can also sanitize. Iodophors are very stable, have a much longer shelf life than hypochlorites, and work at low concentrations. These sanitizing compounds are easy to measure and dispense, and penetrate dirt well. They prevent films and spotting because of their acidity. The temperature of the sanitizing solution should be below 48°C (118°F).

*Quaternary ammonium* compounds work well on floors, walls, equipment, and furnishings in meat and poultry plants. The "quats" work well on porous surfaces because they penetrate well. Quats form a film on surfaces which inhibits bacterial growth. Sanitizers and compounds that contain an acid and a quat sanitizer work best for controlling *Listeria monocytogenes* and mold. Sanitarians may use quats temporarily when they find mold buildup.

*Acid sanitizers* combine rinsing and sanitizing steps. Acid neutralizes the excess alkali from the cleaning residues, prevents alkaline deposits from forming, and sanitizes. Chapter 6 has more information about sanitizers.

## PERSONAL HYGIENE AND WORK HABITS OF EMPLOYEES

### General Instructions

All personnel should have good personal hygiene (see Chap. 4). They should wear clean clothes and stay away from work if they are ill. Only supervisors, managers, and superintendents should have access to cleaning and sanitizing compounds and should be responsible for giving them to cleaning staff. Wrong use of these compounds is expensive, does not result in good cleaning, and could damage employees or equipment. Supervisors should lock the water temperature at 55°C (131°F).

Instructions for portable or centralized high-pressure or foam cleaning systems should be followed. The vendor's recommendations for using cleaning compounds should be followed. (See Chap. 4 for information on safety precautions for cleaning compounds.)

The sanitation supervisor should inspect the entire plant each night after cleaning. All soiled areas should be cleaned again before the regulatory agency inspects the plant in the morning.

Chlorine test papers should be used to check the concentration of sanitizing

solutions if automatic mixing equipment is not available. The test papers come in vials of 100 strips each and are available through most cleaning-compound suppliers.

## Recommended Sanitary Work Habits

Workers should:

1. Store personal equipment (lunch, clothing, etc.) in a sanitary place, and always keep storage lockers clean.
2. Wash and sanitize utensils frequently throughout the production shift, and store them in a sanitary container that will not be in contact with floors, clothing, lockers, or pockets.
3. Not allow food to touch surfaces that have not been sanitized. If any food touches the floor or other unclean surface, it should be rinsed thoroughly with drinkable water.
4. Use disposable towels for hands or utensils.
5. Wear clean clothing in production areas.
6. Cover their hair so that it does not contaminate food.
7. Remove their aprons, gloves, or other clothing that may touch the product before using toilets.
8. Always wash and sanitize their hands when leaving the toilet area.
9. Stay away from production areas when they have an infectious disease, infected wound, cold, sore throat, or skin disease.
10. Not use tobacco in any production area.

## HACCP

In the future, the U.S. Department of Agriculture, which regulates the hygienic production of meat and poultry products, may require use of a HACCP program. (Chapter 10 gives more detail about HACCP.)

HACCP does not have to be expensive. The control points are chosen because they are important for keeping the food safe. Design, maintenance, and process control are important control points and are quite inexpensive. Microbial tests are expensive and do not control the process well.

Sanitation staff can develop a simple HACCP flowchart for the meat or poultry production line. The flowchart includes a long sequence of events with steps that are difficult or impossible to control. Sanitation employees can identify the factors that affect the hazards of each step and decide on critical control points.

## Components of Meat and Poultry Production for HACCP Analysis

***Livestock and poultry production.*** It is possible to raise animals in a specific pathogen-free (SPF) environment to reduce contamination. Farmers can also give animals cultures of good bacteria to compete with pathogens and keep their numbers down.

***Farm environment.*** The farm (pastures, streams, manure, etc.) helps recycle excreted pathogens in a cycle of infection, excretion, and reinfection. Sanitation practices can improve hygiene in this area.

***Transportation.*** Transporting live animals is stressful and may cause animals carrying pathogens to spread these microorganisms. It is challenging to keep transportation sanitary to reduce contamination in the processing plant.

***Animal-holding pens.*** Stress in animal-holding pens can cause changes in the balance of microbes in the intestines so that the animals shed more *Salmonella* spp. in their feces. Showering the animals can reduce the amount of stress and contamination.

***Hide, pelt, hair, or feather removal.*** Meat animals have protective coats that often contain harmful microorganisms. The industry needs better procedures and equipment to remove the outer layers without contaminating the meat.

***Removal of internal organs.*** Intestines can spill and internal organs can burst, releasing microbes onto the meat. A series of water or sanitizer sprays can reduce contamination of poultry or red-meat carcasses. Spraying does not completely remove microorganisms and can spread them over the carcass.

***Inspection.*** A meat inspector can contaminate dressed carcasses with his or her hands or a knife. The inspector should sanitize the knife and his or her hands.

***Chilling.*** Air temperature, air movement, relative humidity, and filtering air all affect the speed of chilling. Keeping the carcass surface dry helps stop microorganisms from growing. Trimming the neck flap area of poultry carcasses after chilling can reduce the number of microbes.

***Further processing.*** Processors should not allow chilled carcasses and cuts of meat to get warm. Processing equipment should have a hygienic design, and should be sanitized before use. Any ingredients added to processed meats should be safe and wholesome.

***Packaging.*** Proper packaging protects the food product from contamination. Proper storage temperatures are also important.

***Distribution.*** Distribution must be fast, clean, and cold. Vehicles should be inspected for cleanliness and temperature control.

## Critical Control Points for Beef Production

The following critical control points (CCPs) have been suggested by the Meat and Poultry Group of the National Committee on Microbiological Criteria for Foods for reducing the risk of pathogens on fresh beef:

*CCP1-Skinning:* The hide is very contaminated, and there is no good way to remove soil from the live animal before slaughter. Therefore, it is important to skin the animal to cut down on cross-contamination from the hide to the carcass.

*CCP2-Prewash:* This step should remove most of the contamination from contact with the hide and minimize attachment of microbes to the carcass.

*CCP3-Bacterial wash:* Bactericidal compounds (e.g., acetic acid) reduce contamination with microbes, including pathogens from the animal's intestines.

*CCP4-Evisceration:* Animals hold large numbers of pathogens in their intestines. The best way to make sure these pathogens do not contaminate the carcass is to teach operators how to remove the intestines and internal organs without bursting them.

*CCP5-Final wash:* This bactericidal rinse reduces the number of microbes on the carcass and minimizes the number of pathogens that are carried into the rest of the processing and packaging process.

*CCP6-Chill:* Rapid chilling controls the growth of pathogens.

*CCP7-Storage:* Strict temperature control and daily cleaning and sanitizing of equipment prevent pathogens from growing. The population of pathogens at this point shows how well the previous CCPs have controlled contamination. Maintaining the product temperature below 7°C (45°F) is a CCP to prevent growth of pathogens during packaging and distribution.

## SANITATION PROCEDURES

It is a good idea to have written cleaning instructions for each operation. Written procedures are especially useful when there is a change in supervisor and for training new employees. More mechanized processes have more detailed and complicated cleaning methods. Written details of cleaning operations should be posted in the plant.

Before deciding on a cleaning procedure, sanitarians must understand how all production and cleaning equipment operates. Here are some examples of cleaning procedures for various areas of a plant. These examples are only guidelines. Sanitarians need to adapt cleaning procedures to the particular area and process.

### Livestock and Poultry Trucks

***Frequency.*** After hauling each load.

***Procedure***

1. Immediately after removing livestock or poultry from trucks, scrape and remove all manure.
2. Clean the truck beds, wheels, and frame by washing down the racks, floors, and frames with water to completely remove all manure, mud, and other debris. Completely disinfect with a quaternary ammonium sanitizer spray, or clean and sanitize in one step by spray cleaning with an alkaline detergent sanitizer.

## Livestock Pens

***Frequency.*** As soon as possible after removing each lot.

***Procedure***

1. After taking the livestock from each pen, clean the manure from the floors and walls, and remove it from the plant.
2. Every 4 months, scrape all dried manure and loose whitewash off the gates and partitions. Sweep cobwebs from the ceilings, and whitewash the inside of the pens. Mix an acid-type sanitizer with the whitewash.
3. If any animals have an infectious disease, quarantine the diseased animals and destroy them separately from the healthy livestock. Remove the manure from the pen area (use a hose if necessary), and disinfect the pens by spraying with a quaternary ammonium sanitizer.

A general cleaning procedure for slaughter and processing areas includes:

1. Removing general debris
2. Prerinsing and wetting
3. Applying the cleaning compound
4. Rinsing
5. Inspecting
6. Sanitizing
7. Preventing recontamination

The first step saves time and water and cuts down the load that enters the sewage system. Removing debris also means less splashing of large particles during the second step.

## Slaughter Area

***Frequency.*** Every day. Debris should be removed several times during the production shift.

***Procedure***

1. Pick up all large pieces of debris and put into trash containers.
2. Cover all electrical connections with plastic sheeting.
3. Quickly prerinse all soiled areas with water at 50 to 55°C (122–131°F). Start with the ceiling, walls, and the top of the equipment, and keep debris moving down to the floor. Do not spray water directly onto motors, outlets, and electrical cables.
4. Apply an alkaline cleaner using a centralized or portable foam system and water at 50 to 55°C (122–131°F). The system should reach all framework, undersides, and other difficult-to-reach areas. Make sure the foam is in contact with the surfaces for 5 to 20 minutes before rinsing. Foam requires less labor, but high-pressure equipment is better for hard-to-reach areas and for removing *Listeria monocytogenes.*

5. Rinse ceilings, walls, and equipment no more than 20 minutes after applying the cleaning compound. Again, start at the ceiling and work downwards, and use water at 50 to 55°C (122–131°F).
6. Inspect all equipment and surfaces, and reclean if necessary.
7. Apply an organic sanitizer to all equipment using a centralized or portable sanitizing unit. If a chlorine sanitizer is used, the solution should be at least 50 parts per million (ppm) of chlorine.
8. Remove, clean, and replace drain covers.
9. Put a white edible oil on surfaces that may rust or corrode.
10. Clean specialized equipment according to the manufacturer's recommendations (if available).
11. Require maintenance workers to carry a sanitizer, and sanitize where they have worked so that equipment is not contaminated during maintenance and setup.

## Poultry Mechanical Eviscerators

***Frequency.*** Daily. Use a continuous or intermittent sanitizer spray to reduce contamination.

***Procedure***

1. Pick up all large pieces of debris and put into trash containers.
2. Cover electrical connections with plastic sheeting.
3. Quickly prerinse the equipment with water at 50 to 55°C (122–131°F).
4. Apply an alkaline cleaner using a centralized or portable foam system and water at 50 to 55°C (122–131°F). Expose the equipment to the cleaner for 10 to 20 minutes, and then rinse with water at 50 to 55°C (122–131°F).
5. Inspect all areas and reclean if necessary.
6. Apply a solution of chlorine at 200 ppm (or other organic sanitizer) using a centralized or portable sanitizing unit.
7. Use sanitizers to avoid contamination during maintenance as before.

## Receiving and Shipping Area

***Frequency.*** Daily.

***Procedure***

1. Cover all electrical connections, scales, and food product with plastic sheeting.
2. Quickly rinse the walls and floors with high-pressure water at 50 to 55°C (122–131°F). Begin rinsing the walls at the top, and rinse dirt down towards the floor. This prerinse removes heavy soil and wets the surfaces.
3. Apply an acid cleaner (55°C [ 131°F] or less) using a slurry or foam gun. Use high pressure (i.e., 25–70 kg/cm$^2$ [360–1,000 lb/in$^2$] and 7.5 to 12 L/min [2–3 gal/min] at the wand).

4. Apply the cleaning compound for 20 minutes, then rinse using high-pressure water at 50 to 55°C (122–131°F).
5. Remove, clean, and replace drain covers.
6. Put all hoses away.

## Processed-Products Area

*Frequency.* Daily.

*Procedure*

1. Take all equipment apart and place the parts on a table or rack.
2. Pick up large pieces of meat and other debris and put into trash containers
3. Cover all electrical connections with plastic sheeting.
4. Prerinse all surfaces with water at 50 to 55°C (122–131°F). Start at the top of processing equipment, and wash debris down to the floor. Do not spray motors, outlets, and electrical cables.
5. Apply an alkaline cleaner using a centralized or portable high-pressure, low-volume system using water at 50 to 55°C (122–131°F). Make sure the system reaches all framework, tables, the undersides of equipment, and other difficult-to-reach areas. Soak the equipment in the cleaning solution for 5 to 20 minutes. Foam can also be used, although it does not penetrate well.
6. Rinse using water at 50 to 55°C (122–131°F). Rinse one side of each piece of equipment at a time.
7. Inspect all equipment surfaces and reclean if necessary.
8. Apply an organic sanitizer to all clean equipment using a centralized or portable sanitizing unit.
9. Remove, clean, and replace drain covers.
10. Use a white edible oil on surfaces that may rust or corrode.
11. Be careful not to contaminate equipment during maintenance.

## Fresh-Product Processing Areas

*Frequency.* Daily.

*Procedure*

1. Take all equipment apart and place the parts on a table or rack.
2. Remove large pieces of debris from equipment and the floor and put into trash containers.
3. Cover mixer, packaging equipment, motors, outlets, scales, controls, and fat analysis equipment with plastic sheeting.
4. Quickly prerinse all soiled surfaces with water at 50 to 55°C (122–131°F) to wash off large pieces of debris and to wet the surfaces. Hose debris towards a floor drain.
5. Apply an alkaline cleaner using centralized or portable high-pressure, low-volume cleaning equipment and water at 50 to 55°C (122–131°F). Or apply the

cleaning compound as a foam, gel, or slurry. Make sure the cleaning compound covers the whole area, including equipment, floors, walls, and doors.
6. Rinse the area and equipment within 20 to 25 minutes after applying the detergent.
7. Inspect the area and equipment and reclean if necessary.
8. Remove, clean, and replace drain covers.
9. Sanitize all clean equipment with an organic sanitizer using a centralized or portable sanitizing unit.
10. Use a white edible oil on surfaces that may rust or corrode.
11. Be careful not to contaminate equipment during maintenance.

## Processed-Products Packaging Area

***Frequency.*** Daily.

***Procedure***

1. Take all equipment apart and place the parts on a table or a rack.
2. Remove large pieces of debris from equipment and floors and put in trash containers.
3. Cover packaging equipment, motors, outlets, scales, controls, and other equipment with plastic sheeting.
4. Prerinse all soiled surfaces with water at 50 to 55°C (122–131°F) to remove heavy soil and to wet the surfaces. Hose debris towards a floor drain.
5. Apply an alkaline cleaner using centralized or portable high-pressure, low-volume cleaning equipment and water at 50 to 55°C (122–131°F). Or use a foam, gel, or slurry to apply the cleaning compound. Make sure the cleaning compound covers the entire area, including equipment, floors, walls, and doors.
6. Rinse the area and equipment within 20 to 25 minutes after applying the cleaning compound.
7. Inspect the area and equipment and reclean if necessary.
8. Remove, clean, and replace drain covers.
9. Sanitize the clean equipment with an organic sanitizer using a centralized or portable sanitizing unit.
10. Use a white edible oil on surfaces that may rust or corrode.
11. Be careful not to contaminate equipment during maintenance.

## Brine-Curing and Packaging Area

***Frequency.*** Daily.

***Procedure***

1. Pick up all large pieces of debris and put in a trash container.
2. Cover all electrical connections, scales, and food products with plastic sheeting.

3. Prerinse the area and equipment with water at 50 to 55°C (122–131°F).
4. Put an acid cleaner in the shrink tunnel (if used), and circulate for about 30 minutes while prerinsing.
5. Rinse the shrink tunnel before applying the detergent.
6. Put all debris from prerinsing in a trash container.
7. Apply an alkaline cleaner using a foam or slurry cleaning system and water at 50 to 55°C (122–131°F).
8. Rinse with water at 50 to 55°C (122–131°F) no more than 20 minutes after applying the detergent.
9. Inspect the area and equipment and reclean if necessary.
10. Remove, clean, and replace the drain covers.
11. Sanitize all clean equipment using an organic sanitizer and a central or portable system.
12. Use white edible oil on parts that may rust or corrode.
13. Be careful not to contaminate equipment during maintenance.

## Smokehouses

***Frequency.*** After each smoke period.

***Procedure***

1. Pick up large pieces of debris and put in trash containers.
2. Apply an alkaline cleaning compound recommended for cleaning smokehouses using a centralized or portable foam system. Use a high-pressure unit where foam cannot penetrate. Figure 12.4 shows a unit for cleaning a smokehouse.
3. Rinse the area no more than 20 to 30 minutes after applying the cleaning compound. Start by rinsing the ceiling and walls, and work all debris down to the floor drain.
4. Inspect all areas and reclean if necessary.
5. Apply an iodophor or quaternary ammonium sanitizer using a centralized or portable sanitizing unit around the entry area to cut down on air contamination.

## Smoke Generator

***Frequency.*** Depends on amount of use.

***Procedure***

*a. Filter*

1. Soak the filter in an alkaline cleaning solution
2. If the filter has mineral deposits, cut the frame apart and clean each leaf. Reweld the frame after cleaning and avoid warping it.




**FIGURE 12.4.** A portable atomizer that covers all stainless-steel surfaces in a smokehouse and reduces cleaning time by 60 to 75%. *(Courtesy of Birko Chemical Corporation.)*

*b. Wash Chamber*

1. Take apart the duct connecting the smoke generator to the house.
2. Clean the soot and ash out of the chamber below the filter.
3. Clean the duct and chamber surface until the metal shows.

## Wire Pallets and Metal Containers

***Frequency.*** Before use.

***Procedure***

1. Use high-pressure water at 50 to 55°C (122–131°F) or cooler as a prerinse.
2. Apply an alkaline cleaner using a foam unit. If foam is not available, use a high-pressure, low-volume unit. Do not spray too many containers at one time. Be careful to rinse the containers before the cleaning compound dries.
3. Rinse with a high-pressure spray of water at 50 to 55°C (122–131°F).
4. Inspect all containers and reclean if needed.

### Offices, Locker Rooms, and Rest Rooms

***Frequency.*** Daily for offices, at least every other day for lockerrooms and rest rooms.

*Procedure*

1. Cover electrical connections with plastic sheeting.
2. Clean areas with a foam or high-pressure unit (or scrub and mop).
3. No more than 20 minutes after applying the cleaning compound, rinse with water at 50 to 55°C (122–131°F).
4. If these steps do not clean dirty areas or if drains are not present, hand-scrub with scouring pads.

## TROUBLESHOOTING TIPS

*Discolored floors:* Lighten darkened concrete floors using a bleach solution (soak for at least 30 minutes). Finish cleaning the floor using a mechanical scrubber.

*White film buildup on equipment:* Too much cleaning compound, poor rinsing, or hard water cause this film. Do not use too much cleaning compound, rinse thoroughly, and use soft water.

*Conveyor wheels freezing:* The cleaning water is probably too hot. Wheels lose lubricant when the temperature reaches 90°C (194°F). The cleaning solutions should not be hotter than 55°C (131°F). Relubricate the conveyor wheels (follow the manufacturer's directions).

*Sewer lines plugged:* Clean sediment bowls daily, and do not flush floor sweepings into sewer pipes.

*Yellow protein buildup on equipment:* The water may be too hot. Brush away the organic material that builds up every day. If soil has been cooked onto equipment for a long time, rub with steel wool to remove the soil.

*Do not spray:* liver slicers, cube steak machines, electronic scales, patty machines, electrical outlets, motors, or equipment with open connections, wrapping film or containers, and wrapping units. Cover all outlets with polyethylene bags, and cover electrical equipment with waterproof drop cloths. Follow the manufacturer's recommendations when cleaning this equipment.

## SUMMARY

- A good cleaning system can cut labor costs in meat and poultry plants by as much as 50%.
- *L. monocytogenes* is a common pathogen in meat- and poultry-processing plants.
- The optimal cleaning system depends on the type of soil and the type of equipment. High-pressure, low-volume cleaning equipment works best for removing heavy organic soil, especially in areas that are difficult to reach. But foams, slurries, and gels are quicker and easier to apply. CIP systems are expensive and do not work well in meat and poultry plants, except in large storage containers.
- HACCP is a good way to make sure that meat and poultry products are safe to eat.

- Operators of meat and poultry plants use acid cleaning compounds to remove mineral deposits and alkaline cleaning compounds to remove organic soils. Chlorine compounds are the best and cheapest sanitizers. But iodine compounds are less corrosive and irritating, and quaternary ammonium sanitizers have a longer-lasting effect.
- Operators should have written cleaning procedures for each area of the plant.

## BIBLIOGRAPHY

Daun, H., Solberg, M., Franke, W., and Gilbert, S. 1971. Effect of oxygen-enriched atmospheres on storage quality of packaged fresh meat. *J. Food Sci.* 36:1001.

Douglas, G. S. 1987. Listeria and food processing. *Technics/Topics* 11:2.

Doyle, M. P. 1987. Low-temperature bacterial pathogens. *Proc. Meat Ind. Res. Conf.*, p. 51. American Meat Institute, Washington, D.C.

Ledward, D. A. 1970. Metmyoglobin formation in beef stored in carbon dioxide enriched and oxygen depleted atmospheres. *J. Food Sci.* 35:33.

Marriott, N. G. 1994. *Principles of Food Sanitation,* 3d ed. Chapman & Hall, New York.

Marriott, N. G. 1990. *Meat Sanitation Guide II.* American Association of Meat Processors and Virginia Polytechnic Institute and State University, Blacksburg.

Price, J. F., and Schweigert, B. S. 1987. *The Science of Meat and Meat Products,* 3d ed. Food & Nutrition Press, Westport, Conn.

Roback, D. L., and Costilow, R. M. 1961. Role of bacteria in the oxidation of myoglobin. *Appl. Microbiol.* 9:529.

Shapton, D. A., and Shapton, N. F., eds. 1991. Buildings. In *Principles and Practices for the Safe Processing of Foods,* p. 37. Butterworth-Heinemann, Oxford.

Smulders, F. J. M. 1987. Microbial contamination and decontamination. *Proc. Meat Ind. Res. Conf.*, p. 29. American Meat Institute, Washington, D.C.

Solberg, M. 1968. Factors affecting fresh meat color. *Proc. Meat Ind. Res. Conf.*, p. 32. American Meat Institute Foundation, Chicago.

### STUDY QUESTIONS

1. Why doesn't refrigeration control *L. monocytogenes?*
2. List six steps to control *L. monocytogenes.*
3. Give five reasons why good sanitation is important in meat- and poultry-processing facilities.
4. What are the basic steps of a cleaning procedure?
5. Give three examples of CCPs.
6. Why are written sanitation procedures important?

### TO FIND OUT MORE ABOUT MEAT AND POULTRY SANITATION

1. Call the United States Department of Agriculture (USDA) Meat & Poultry Hotline, (1-800) 535-4555 for answers to questions about handling meat and poultry, how to tell if it's safe to eat, and how to read meat and poultry labels. The hotline is staffed from 8 a.m.

to 5 p.m. (EST), Monday to Friday. You can also write to: The Meat and Poultry Hotline, USDA-FSIS, Room 1163-S, Washington, DC 20250.

2. Call the toll-free number on a package of processed meats, and ask if the manufacturer has information on food safety of their products.
3. Contact the National Cattlemen's Beef Association, (312) 467-5520, 444 North Michigan Avenue, Chicago, IL 60611, and ask for information on food safety and sanitation for red-meat products.



### CHOOSING THE RIGHT CRITICAL CONTROL POINT AND THE RIGHT CRITICAL LIMIT

For HACCP to work, it is important to find out what the critical control points (CCPs) are, i.e., where the meat becomes most contaminated during processing. Each CCP then needs a critical limit, i.e., whatever is needed to keep the contamination low enough to be acceptable at that point.

This study found that inspecting lamb carcasses is not a good way to tell whether a carcass is heavily contaminated with microbes. Carcasses that had visible pieces of wool, skin, feces, or other intestinal matter were not always carrying more microbes than carcasses that looked clean. But carcasses from animals that had long wool or that had been washed before slaughter carried more microbes than animals with short wool or that had not been washed. Therefore, checking the length of the wool on the lambs when they arrive at the slaughterhouse and not prewashing the lambs are more important than checking carcasses on the line for obvious contamination.

This study shows that sanitarians need to do a careful hazard analysis and use microbiological testing when they set up the HACCP system, to make sure that the things they decide to monitor are really the most important points in the process where the food product could be contaminated.

*Source:* Biss, M.E., and Hathaway, S.C. 1995. Microbiological and visible contamination of lamb carcasses according to preslaughter presentation status: implications for HACCP. *J. Food Protection.* 58:776.